

CSE340: Computer
Architecture
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Section: 04

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1.

```
if (A[3] + B[2] >= A[5]) {  
    if ((A[4] >> 2) == (B[3] & 0xFF)) {  
        A[3] = A[3] / 2 + B[1];  
    } else {  
        B[2] = A[4] * 5 - B[2];  
    }  
} else {  
    A[5] = B[3] << 3;  
}
```

RISC-V Code: { Base address of A is in x9 and Base address of B is in x18 }

```
ld x28, 24(x9)  
ld x29, 16(x18)  
add x30, x28, x29  
ld x31, 40(x9)  
blt x30, x31, Level_1  
ld x7, 32(x9)  
srli x7, x7, 2  
ld x6, 24(x18)  
andi x6, x6, 0xFF
```

```
bne x7, x6, Level_2  
srli x30, x28, 1  
ld x31, 8(x18)  
add x30, x30, x31  
sd x30, 24(x9)  
beq x0, x0, end  
Level_2: ld x7, 32(x9)  
srli x30, x7, 2  
add x30, x30, x7  
sub x31, x30, x29
```

```

sd x31, 16(x18)
beq x0, x0, end
Level_1: ld x30, 24(x18)
          slli x30, x30, 3
          sd x30, 40(x9)
End:

```

2.

```

if (A[3] + B[2] > A[5]) {
    if ((A[4] << 1) <= (B[A[3]])) {
        A[3] = A[3] / 2 + B[1];
    } else {
        B[2] = A[4] * 5 - B[2];
    }
} else {
    A[5] = B[3] >> 3;
}

```

RISC-V Code: { Base address of A is in x9 and Base address of B is in x18 }

```

ld x28, 24(x9)
ld x29, 16(x18)
add x30, x28, x29

```

```

ld x31, 40(x9)
ble x30, x31, Level_1
ld x7, 32(x9)

```

```

slli x7, x7, 1
slli x30, x28, 3
add x30, x30, x18
ld x31, 0(x30)
bgt x7, x31, Level_2
slli x30, x28, 1
ld x31, 8(x18)
add x30, x30, x31
sd x30, 24(x9)
beq x0, x0, End
Level_2: ld x7, 32(x9)
        slli x30, x7, 2
        add x30, x30, x7
        sub x31, x30, x29
        sd x31, 16(x18)
        beq x0, x0, End
Level_1: ld x30, 24(x18)
        slli x30, x30, 3
        sd x30, 40(x9)
End:

```

3.

```
for (i = 0; i < 10; i++) {  
    A[4] = A[i] ^ B[i];  
}
```

RISC-V Code: { Base address of A is in x9 and Base address of B is in x18 }

```
add x5, x0, x0
```

```
addi x6, x0, 10
```

```
Loop: bge x5, x6, End
```

```
    slli x7, x5, 3
```

```
    add x30, x9, x7
```

```
    add x31, x18, x7
```

```
    ld x28, 0(x30)
```

```
    ld x29, 0(x31)
```

```
    xor x30, x28, x29
```

```
    sd x30, 32(x9)
```

```
    addi x5, x5, 1
```

```
    beq x0, x0, Loop
```

```
End:
```

4.

```
for (i = 10; i >= 0; i--) {  
    if ((A[i] & 0xF) == (B[i] | 0x3)) {  
        A[B[i]] = A[i] ^ B[i];  
    }  
}
```

RISC-V Code: { Base address of A is in x9 and Base address of B is in x18 }

add x5, x0, 10

addi x6, x0, x0

Loop: blt x5, x6, End

add x31, x9, x5

add x32, x18, x5

ld x28, 0(x31)

ld x29, 0(x32)

andi x30, x28, 0xF

ori x32, x29, 0x3

bne x30, x32, skip

xor x30, x28, x29

add x7, x31, x29

sd x30, 0(x7)

beq x0, x0, Loop

skip: addi x5, x5, -1

beq x0, x0, Loop

End:

5.

Memory Location	Code	Line Number	Machine Code
	ADDI X ₅ , X ₀ , 5	1	
	ADDI X ₆ , X ₀ , 1	2	
	ADDI X ₂₅ , X ₀ , 0	3	
	Loop: BLT X ₅ , X ₆ , loopBreak	4	_____XXX_____XXXXXXXX
	ADDI X ₂₅ , X ₂₅ , 1	5	
#7080	ADDI X ₅ , X ₅ , -1	6	
	BEQ X ₀ , X ₀ , Loop	7	_____XXX_____XXXXXXXX
	loopBreak:	8	

- a) What is the value of **PC** while executing line2? Answer:
- _____
- b) Fill up the machine codes corresponding to line4 and line7 in the table above.

Answer:

a) Here, Total lines = 8

Memory location of 6th line = 7080

Each line contains 4 byte or 32 bit.

So, value of pc of line 2 = $7080 - (6-2) \times 4$
= 7064

b) Line 4 = Loop: BLT $X_5, X_6, \text{loopBreak}$

$$rs1 = (5)_{10} = (00101)_2$$

$$rs2 = (6)_{10} = (00110)_2$$

$$\text{line jump} = 8 - 4 = 4$$

$$\text{memory pc update} = 4 \times 4 = 16$$

$$\text{forward} = (16)_{10} = (0000000010000)_{16}$$

$$\text{imm}[12] = 0$$

$$\text{imm}[10:5] = 000000$$

$$\text{imm}[4:1] = 1000$$

$$\text{imm}[11] = 0$$

0	000000	00110	00101	xxx	1000	0	XXXXXXXX
imm[12]	imm[10:5]	rs2	rs1	func3	imm[4:1]	imm[11]	opcode

Line 7 = BEQ X_0, X_0, Loop

$$rs1 = (0)_{10} = (00000)_2$$

$$rs2 = (0)_{10} = (00000)_2$$

$$\text{line jump} = 4 - 7 = -3$$

$$\text{memory pc update} = -3 \times 4 = -12$$

$$\text{forward} = (-12)_{10} = (1111111110100)_{16}$$

$$\text{imm}[12] = 1$$

$$\text{imm}[10:5] = 111111$$

$$\text{imm}[4:1] = 1010$$

$$\text{imm}[11] = 1$$

1	111111	00000	00000	xxx	1010	1	XXXXXXXX
imm[12]	imm[10:5]	rs2	rs1	func3	imm[4:1]	imm[11]	opcode

Sign bit
↓

$$\begin{array}{r}
 0000000001100 \\
 1111111110011 \\
 \text{(1's complement)} \\
 1111111110011 \\
 \quad \quad \quad +1 \\
 \hline
 1111111110100 \\
 \text{(2's complement)}
 \end{array}$$

6. For the following RISC-V assembly instructions, determine the corresponding

C/high-level statement:

slli x30, x6, 2

add x30, x10, x30

slli x31, x7, 2

add x31, x11, x31

lw x6, 0(x30)

lw x7, 0(x31)

xor x8, x6, x7

slt x9, x6, x7

add x8, x8, x9

sw x8, 0(x30)

Assumptions:

- Variables m, n, p, q, r are assigned to registers x6, x7, x8, x28, and x29, respectively.

- Base addresses of arrays X and Y are in registers x10 and x11, respectively.

Answer:

slli x30, x6, 2 $\rightarrow x30 = m * 4$

add x30, x10, x30 $\rightarrow x30 = \&X[m]$

```

slli x31, x7, 2  →  $x_{31} = n * 4$ 
add x31, x11, x31 →  $x_{31} = \&Y[n]$ 
lw x6, 0(x30) →  $m = X[m]$ ;
lw x7, 0(x31) →  $n = Y[n]$ ;
xor x8, x6, x7 →  $p = m \wedge n$ ;
slt x9, x6, x7 →  $x_9 = (m < n) ? 1 : 0$ ;
add x8, x8, x9 →  $p = (m \wedge n) + (m < n)$ ;
sw x8, 0(x30) →  $X[m] = p$ 

```

Equivalent C code:

```

m = X[m];
n = Y[n];
p = (m ^ n) + (m < n);
X[m] = p;

```

Final code:

```

X[m] = (X[m] ^ Y[n]) + (X[m] < Y[n]);

```